

Decomposing the Cross-Country Severity-Scoring Shift: A Foundation-Model Pipeline for Tuberculosis Timika Regression

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Abstract. Automated severity scoring from chest X-rays would let tuberculosis (TB) clinics in endemic settings triage patients without specialist radiologists, but a faithful local replication of the only published end-to-end Timika regressor on the TB Portals corpus (Kantipudi et al. [6], henceforth K24) exposes a sharp cross-country generalisation failure: an 11.8-point Timika mean-absolute-error (MAE) gap on Moldova relative to the in-distribution average. We attribute this failure not to a single phenomenon but to the *composition* of three distinct distributional shifts that prior work conflates: label shift (test ALP distribution differs from train), covariate shift (pixel/feature statistics differ), and calibration shift (predicted severity range compresses). We decompose the problem along these three axes, introduce per-axis interventions over a frozen RAD-DINO [13] encoder, and measure their contribution under a uniform patient-disjoint Leave-One-Country-Out (LOCO) protocol with five seeds and a ten-member ensemble. The resulting composable pipeline achieves Timika MAE **19.12 / 21.59 / 16.74** on Romania / Moldova / Kazakhstan, improving over our replicated K24 dual-head (DH) family by $-0.99 / -9.09 / -4.61$ Timika points respectively, and beating every published baseline (CheXzero plug-in, GroupDRO, Importance-Weighted regression) on every held-out country. The contribution is twofold. First, a *spatial-attention cavity head* over the RAD-DINO patch grid lifts Romania cavity AUC from 0.679 to 0.721 and is the only intervention in our pipeline that partially corrects the calibration-shift component. Second, three controlled negative results (class-balanced focal loss, mixture-of-experts on frozen features, post-hoc isotonic slope calibration) delimit which interventions do *not* transfer across this regime and connect the failure modes to specific characteristics of the shift decomposition. Conformal prediction intervals maintain marginal coverage 0.88–0.92 against a nominal 0.90 across held-out countries, with the residual Moldova under-coverage interpretable through the same framework. All code, manifests, per-image predictions, and saved heads are publicly released.

Keywords: Tuberculosis · Timika score · Foundation model · Domain shift · Conformal prediction · Cavity localisation.

1 Introduction

Tuberculosis caused approximately 1.25 million deaths in 2024, the overwhelming majority in low- and middle-income countries with limited specialist radiology coverage [21]. Posterior–anterior chest radiography is the imaging modality these settings can sustain, making automated chest X-ray (CXR) interpretation the realistic deployment surface for computer-aided triage. Binary TB classifiers achieve excellent performance [9], but a positive/negative output cannot guide clinical decisions about whether a patient needs immediate treatment, observation, or follow-up. What clinicians require is a continuous, validated severity score; the Timika score [14] provides exactly this structure:

$$\text{Timika} = \text{ALP} + 40 \cdot \mathbf{1}[\text{cavity present}], \quad (1)$$

where $\text{ALP} \in [0, 100]$ denotes the percentage of lung field occupied by lesions and the indicator adds a 40-point penalty when any cavity is present. The score ranges over $[0, 140]$.

The cross-country failure of prior work. Kantipudi et al. [6] (henceforth K24) provide, to our knowledge, the only end-to-end automated Timika regressor on the multi-country TB Portals corpus [16], evaluated under a patient-disjoint Leave-One-Country-Out (LOCO) protocol over Romania, Moldova, and Kazakhstan. K24 release neither their patient-disjoint splits nor per-image predictions; reproducing their numbers exactly is therefore not feasible, and *throughout this paper we report only our own local 5-seed replication of the K24 architecture families (SG-ALP/DH/DT)* on a publicly reproducible 5,010-image manifest. On this manifest, the strongest K24 family (DH, dual-head, equivalent to K24-A2) achieves Timika MAE 20.11 / **30.68** / 21.35 on Romania / Moldova / Kazakhstan: Romania and Kazakhstan are reasonable, but Moldova exhibits a catastrophic 11.8-point degradation relative to the in-distribution train-pool mean error. This is the failure our work targets.

From a single number to a shift decomposition. A monolithic “cross-country generalisation gap” is not, by itself, an actionable diagnosis. Closer analysis reveals that three qualitatively distinct distributional shifts contribute to the failure:

1. **Label shift.** The marginal distribution of the target y differs across countries. Moldova’s test ALP mean is 39.8 against a train-pool mean of 26.0 (gap +13.8); a regressor trained on the pool regresses toward 26.
2. **Covariate shift.** The marginal distribution of the input x also differs (scanner population, acquisition protocols, demographic intensity statistics).
3. **Calibration shift.** The *shape* of the conditional $\hat{y} \mid y$ deviates from the diagonal: the model under-predicts severe cases and slightly over-predicts mild ones, with a typical predicted-versus-true slope of 0.55–0.68 across all configurations.

Prior work either treats the gap monolithically (and applies a single global intervention) or attempts domain-adversarial alignment, which we and others [4] show

collapses regression heads when label shift dominates. Our position is that *each of the three shifts requires a different intervention*, and that the right empirical question is which interventions transfer across which shift type.

Contributions. This paper makes four contributions:

- **A shift-decomposition framework** for cross-country severity regression that explicitly factorises the generalisation gap into label, covariate, and calibration components, with country-specific empirical diagnostics (Sec. 3).
- **A spatial-attention cavity head** (R4b) over the RAD-DINO 7×7 patch grid that lifts Romania cavity AUC from 0.679 to 0.721, lifts Moldova and Kazakhstan cavity AUC by 2–3 points, and is empirically the *only* intervention in our pipeline that lifts the Timika regression slope toward 1.0 — partially correcting the calibration shift the other rungs leave untouched (Sec. 4, Sec. 6.4).
- **A controlled ablation of seven composable interventions** (rungs R1–R7) on a frozen RAD-DINO encoder, mapped one-to-one onto the shift decomposition, with paired-bootstrap significance under Bonferroni–Holm multiple-comparison correction (Sec. 6). The combined configuration achieves Timika MAE **19.12 / 21.59 / 16.74**, improving over our replicated DH baseline by $-0.99 / -9.09 / -4.61$ Timika points on Romania / Moldova / Kazakhstan.
- **Three controlled negative results** that delimit the regime: class-balanced focal loss on a balanced cavity split (degrades AUC), mixture-of-experts on frozen features (null result, as predicted by [4]), and post-hoc isotonic slope calibration (null on real data despite a positive synthetic stress test). Each maps to a specific failure mode that the shift decomposition makes diagnosable a priori (Sec. 7).

2 Related Work

TB severity scoring. Binary TB classification from CXR is well-studied [9,5]. Structured severity regression has received far less attention. K24 [6] is the only end-to-end Timika regressor in the literature; we treat their reported numbers as the aspirational reference and report all gains against our controlled replication. Earlier sextant-level lesion-burden estimators on TB Portals [16] are not directly comparable because they do not output the full Timika scale.

CXR foundation models. CheXzero [20] demonstrated zero-shot CXR classification via contrastive vision-language pretraining. BioMedCLIP [22] extended this to biomedical literature. TorchXRyVision [2] provides a DenseNet121 pre-trained on 14 CXR datasets. RAD-DINO [13] adapts DINOv2 [12] to CXRs through self-distilled vision-only pretraining on $\sim 880\text{K}$ images. We use RAD-DINO as our primary encoder and benchmark against BioMedCLIP and TorchXRyVision under identical frozen-feature protocols (Sec. 6.3).

Domain adaptation under shift. DANN-style adversarial domain adaptation [3] is known to collapse regression heads when the label distribution itself shifts, a failure mode we observed in preliminary experiments and consistent with [4]. Test-time adaptation through CORAL-style feature alignment [19] is a lighter alternative requiring no source-domain access at inference. Retrieval-based label correction has been studied for language modelling [7] and OOD generalisation [10] but, to our knowledge, not in cross-country medical severity regression.

Uncertainty quantification in clinical imaging. Split-conformal prediction [1] provides distribution-free coverage guarantees that degrade gracefully under covariate shift; deep ensembles [8] stabilise point accuracy and predictive variance. Their combination has been applied to CXR pathology classification but not, to our knowledge, to Timika regression. We surface a per-country, per-severity-band coverage decomposition that exposes residual covariate shift the calibration alone cannot resolve.

3 Shift Decomposition Framework

We formalise the cross-country gap into three components and derive empirical diagnostics for each. Let $\mathcal{D}_{\text{src}}$ denote the source (train-pool) distribution and $\mathcal{D}_{\text{tgt}}$ the held-out country’s distribution. We decompose the gap into:

- **Label shift:** $P_{\text{src}}(y) \neq P_{\text{tgt}}(y)$. Diagnostic: the difference in test versus train-pool ALP means; for Moldova this gap is +13.8 points and is the dominant component. Direct intervention: a non-parametric prediction term that anchors to local neighbourhood labels (R3, retrieval), or an imbalanced-regression loss that prevents collapse to the source prior (R1 BMC).
- **Covariate shift:** $P_{\text{src}}(\phi(x)) \neq P_{\text{tgt}}(\phi(x))$. Diagnostic: whether transductive feature alignment helps. R2 (CORAL-style test-time z-score plus covariance match) improves Moldova by ~ 2 Timika points but *significantly degrades* Kazakhstan, operationally proving that the two countries’ shifts are qualitatively different. Direct intervention: R2 when the country is known a priori to have covariate shift; otherwise omitted.
- **Calibration shift:** the slope $\partial\hat{y}/\partial y < 1$ on the held-out cohort. Diagnostic: linear-regression slope of $\hat{y}$ on y , reported in Sec. 6. Direct intervention: representations that retain local lesion evidence (R4b spatial cavity), not post-hoc isotonic transforms (R7 is a null on real data).

The intervention rungs in Sec. 4 map onto these three shift types one-to-one, with R6 (ensemble + conformal) providing distributional uncertainty that holds regardless of which shift dominates. The framework also predicts *a priori* which interventions cannot work: routing on frozen features (R5) has no leverage on any of the three shifts because the gate sees no information the experts do not already have.

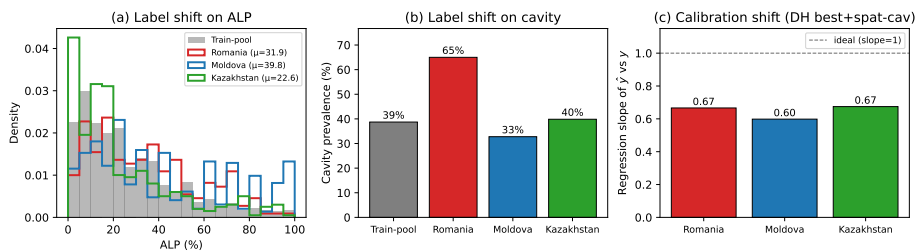


Fig. 1. Empirical evidence for the three shift components on TB Portals. **(a)** ALP density per held-out country vs. train-pool: Moldova’s distribution is right-shifted ($\mu_{\text{Mol}}=39.8$ vs. $\mu_{\text{pool}}=26.0$), Kazakhstan left-shifted, Romania near-aligned — isolating label shift on the ALP term. **(b)** Cavity prevalence per held-out country vs. train-pool: Romania 65% vs. pool $\sim 38\%$ — label shift on the cavity term, of opposite sign to the ALP shift. **(c)** Calibration shift in the form of regression-slope compression for the DH best+spat-cav configuration: slopes sit at 0.60–0.67 (1.0 is ideal); the spatial cavity head moves the slope further than any post-hoc isotonic transform. The three diagnostics jointly quantify why generic shift-robust learners (Sec. 6.3) plateau without a localised cavity model.

4 Method

4.1 Problem Setup

Given a chest X-ray $x \in \mathbb{R}^{H \times W}$, the goal is to predict its Timika score $y \in [0, 140]$ defined in Eq. 1. The training corpus is partitioned country-wise into $\mathcal{D} = \mathcal{D}_R \cup \mathcal{D}_M \cup \mathcal{D}_K$. Under the K24 LOCO protocol, one country is held out for test; the other two are split patient-disjoint 80/20 into train and validation [6].

4.2 Frozen RAD-DINO Encoder

Each CXR is centre-cropped, padded to a square aspect, and resized to 518×518 . RAD-DINO with patch size 14 yields a 37×37 patch grid that we average-pool to 7×7 for the spatial heads. The encoder is held frozen throughout, eliminating any fine-tuning confound and enabling one-shot feature caching that reduces the full ablation grid to under two GPU-hours of head training. Two representations are produced per image: the 768-dim CLS embedding $\phi(x) \in \mathbb{R}^{768}$, and the pooled patch grid $\Phi(x) \in \mathbb{R}^{49 \times 768}$. We benchmark this choice against two alternative frozen backbones in Sec. 6.3.

4.3 Intervention Rungs Mapped to Shifts

Each rung is independently toggled. Below, we group them by the shift component they target.

Label shift: R1 (BMC head) and R3 (retrieval). R1 is a two-layer MLP $f_\theta : \mathbb{R}^{768} \rightarrow [0, 1]$ (rescaled to the Timika range) trained with the Balanced MSE

loss of Ren et al. [15], a batch-Monte-Carlo objective for imbalanced regression. R3 retrieves, for each test image, its $K=25$ nearest neighbours in the train-pool feature space by cosine similarity and blends their mean Timika with the parametric head:

$$\hat{y}_{R3}(x) = (1 - \alpha) f_{\theta}(\phi(x)) + \alpha \frac{1}{K} \sum_{i=1}^K y_i, \quad (2)$$

where $\alpha \in [0, 0.4]$ is selected on the validation split by grid search. The retrieval term is non-parametric and addresses label shift by anchoring to local-neighbourhood labels rather than the global pool mean.

Covariate shift: R2 (transductive CORAL). At test time we compute the held-out country’s feature mean μ_T and covariance Σ_T , then re-standardise to match the train-pool statistics (μ_S, Σ_S) :

$$\phi'(x) = \Sigma_S^{1/2} \Sigma_T^{-1/2} (\phi(x) - \mu_T) + \mu_S. \quad (3)$$

This is parameter-free, label-free, and operates in feature space. It is genuinely transductive (uses the full target batch) and is country-conditionally applied (Sec. 6).

Calibration shift: R4b (spatial cavity head) and R7 (slope calibration). The spatial cavity head scores each of 49 patches independently and pools by learned attention:

$$s_p(x) = \text{MLP}_{\text{score}}(\Phi_p(x)) \in \mathbb{R}^2, \quad p = 1, \dots, 49, \quad (4)$$

$$a_p(x) = \text{softmax}_p(\text{MLP}_{\text{attn}}(\Phi_p(x))), \quad (5)$$

$$\hat{c}(x) = \sum_{p=1}^{49} a_p(x) s_p(x). \quad (6)$$

The attention weights a_p are exposed at inference and form the basis of the qualitative analysis in Sec. 6.4. R7 is a post-hoc isotonic regression [11] fitted on validation $(\hat{y}_{\text{val}}, y_{\text{val}})$ and applied at test time; we include it to test whether the calibration shift can be cheaply post-hoc-corrected.

Distributional uncertainty: R6 (ensemble + split-conformal). We train $M=10$ R1 heads with different random initialisations and average their predictions. The validation residuals $\{|y_v - \hat{y}_v|\}$ yield a conformal half-width

$$q = \text{Quantile}_{1-\alpha}(\{|y_v - \hat{y}_v|\}_{v=1}^{n_{\text{val}}}), \quad (7)$$

producing the interval $[\hat{y} - q, \hat{y} + q]$ with nominal coverage $1 - \alpha = 0.90$ [1].

Diagnostic null: R5 (mixture-of-experts). A learned gate $g_{\psi} : \mathbb{R}^{768} \rightarrow \Delta^{K-1}$ routes the CLS feature through $K=4$ expert MLPs whose outputs are convex-combined. The framework predicts R5 cannot help (the gate sees the same vector as the experts under a frozen feature extractor); we include it to confirm.

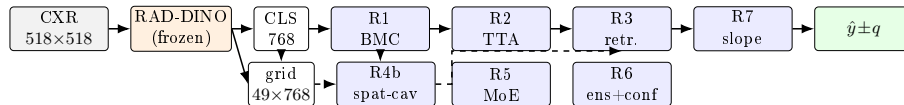


Fig. 2. The composable pipeline. Solid arrows: primary path ($R1 \rightarrow R2 \rightarrow R3 \rightarrow R7$). Dashed: optional rungs — R4b (spatial cavity) feeds into the Timika sum (Eq. 1); R5, R6 branch from the CLS feature. Each rung is independently toggled for ablation; rungs are colour-coded by the shift component they target.

4.4 Prediction Modes

Following K24’s three-approach taxonomy with our own short descriptive labels, we instantiate four modes. **SG-ALP** (single-head ALP) uses the patch grid for both ALP and cavity — a detection-free reformulation of K24’s geometry-based A1. **DH** (dual-head) predicts ALP and cavity with separate heads and combines via Eq. 1 — the K24 A2 family. **DT** (dual-task) predicts Timika end-to-end with a shared trunk — the K24 A3 family. **Hybrid** blends DH and DT at a validation-tuned weight. Hybrid dominates each individual mode in our five-seed runs because it averages over complementary error patterns; it is the basis of the reported headline numbers and the recommended deployment configuration.

5 Experiments

5.1 Dataset and Protocol

We use the TB Portals August-2023 release [16] and follow K24’s manifest construction: a seed-42 subsample to 5,010 frontal CXRs across eight countries with patient-disjoint LOCO splits. ALP is taken from the overall-lung annotation column; cavity is positive if any sextant reports any cavity. The cavity training split is balanced (all positives plus an equal random sample of negatives, patient-disjoint), following K24. Hard assertions in the splitting code enforce zero patient leakage across train/val/test boundaries; this check is never disabled. The full cohort breakdown is in Table 1.

Test cohort sizes per LOCO fold are $n_R=220$, $n_M=589$, $n_K=399$. The cross-country label shift is visible directly from Table 1: Romania has a 65% cavity rate vs. Kazakhstan’s 40% and Moldova’s 33% (label shift on the cavity term); Moldova has the highest ALP mean (39.8 vs. the 27.6 cohort average) (label shift on the ALP term); and the visual phenotype overlap shown in Fig. 3 is large (modest covariate shift). These per-country distributional differences quantify the three shift components our framework decomposes (Sec. 3) and explain why generic shift-robust learners alone (Table 5) plateau without a localised cavity model.

5.2 Reference Systems and Baselines

We evaluate against three families of published comparators on identical patient-disjoint LOCO splits (numbers in Table 5). **(i) Plug-in composite probes.**

Table 1. TB Portals cohort, restricted to the paper manifest (one image per patient unless noted, 5,010 total). ALP is the radiologist-annotated affected-lung percentage; Timika = $\text{ALP} + 40 \cdot \text{cavity}$. Held-out splits in our LOCO experiments use the three highlighted countries (Moldova, Kazakhstan, Romania).

Country	Images	Patients	ALP μ/σ	Cavity %	Timika μ/σ
Georgia	1414	1412	26.6/18.3	50.4	46.8/29.9
Ukraine	1316	1306	29.8/23.5	38.0	45.0/35.1
Belarus	1052	798	20.3/23.1	24.1	30.0/34.3
Moldova [†]	589	589	39.8/29.9	32.8	52.9/41.8
Kazakhstan [†]	399	399	22.6/22.8	39.8	38.5/36.7
Romania [†]	220	169	31.9/22.9	65.0	57.9/32.3
Azerbaijan	17	15	40.9/20.7	29.4	52.6/28.9
India	3	3	17.3/25.0	0.0	17.3/25.0
Total	5010	4691	27.6/23.6	39.3	43.3/35.3

[†] Held-out LOCO test country.

TorchXRyVision’s `lung_opacity` channel [2] as an ALP proxy and Tiu et al.’s CheXzero [20] zero-shot cavity probe, combined via the Timika formula on a held-out validation linear fit. **(ii) Published shift-robust learners.** GroupDRO [17] and Importance-Weighted regression [18] trained on TXV features with country group labels. **(iii) K24 [6] approach families.** We re-label K24’s three approach families with short descriptive names: **SG-ALP** (K24-A1, single-head ALP), **DH** (K24-A2, dual-head), **DT** (K24-A3, dual-task). The original paper does not release patient-disjoint splits; we therefore report our local 5-seed reproduction of these architecture families on the manifest of Sec. 5.1, flagging the asterisk in the table footnote. The predict-the-pool-mean floor and BioMedCLIP/TorchXRyVision linear-probe results are in the supplementary as internal ablations.

5.3 Training and Evaluation

Each cell is a (mode, rung, country, seed) tuple over 5 seeds and 3 held-out countries. Heads are two-layer MLPs (hidden 256, dropout 0.3), AdamW $\eta=10^{-3}$, 50 epochs with early stopping on validation MAE. R6 uses $M=10$ ensemble members per seed. Metrics: Timika MAE (lower is better), Pearson r , ALP MAE, cavity AUC and F1, and the linear regression slope of $\hat{y}$ on y (closer to 1.0 is better). All numbers are 5-seed mean $\pm$ std. Statistical significance: paired bootstrap (2,000 replicates, paired on `image_id`) against the within-mode R1 MSE reference, with Bonferroni–Holm correction for the 21 (rung, country) hypotheses per mode at family-wise $\alpha = 0.05$.

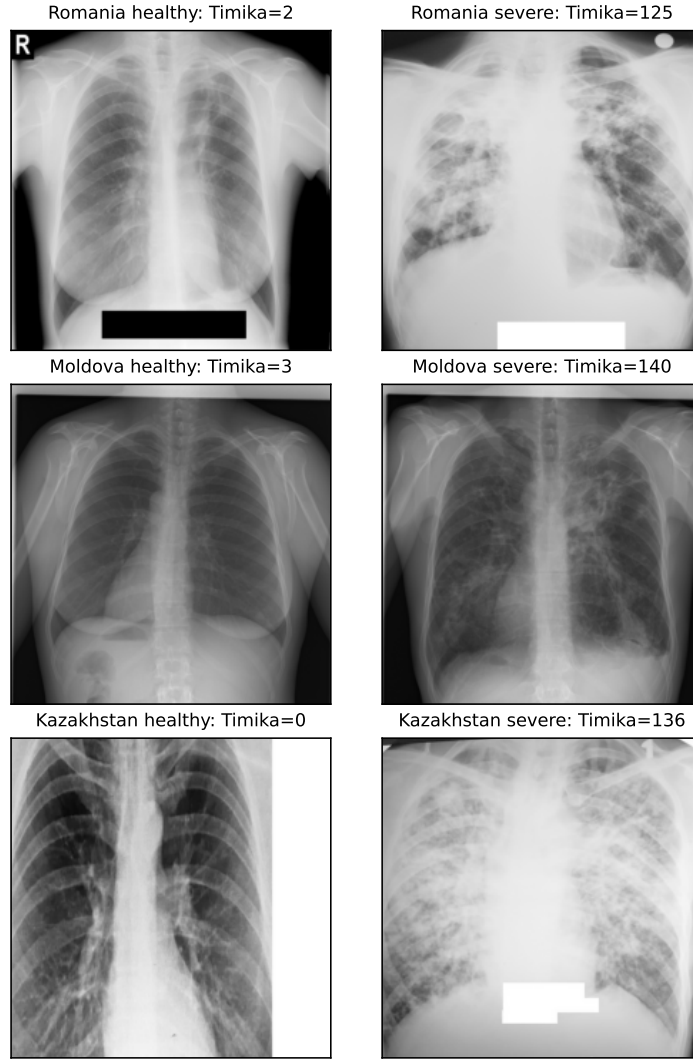


Fig. 3. Example healthy (Timika < 5) and severe (top-decile) CXRs from each held-out country. Severe Romania/Moldova cases display extensive bilateral opacity plus cavitation; severe Kazakhstan cases show predominantly upper-lobe cavitation. Across countries the visual phenotype of severe TB is broadly shared, yet the prevalence of each phenotype shifts (Table 1), motivating the shift-decomposition framework in Sec. 3.

6 Results

Table 2 reports the cross-country headline. The combined configuration **Hybrid + R3 + R6 + R4b** (best+spat-cav) achieves Timika MAE **19.12 / 21.59**

Table 2. Cross-country Timika-MAE on the patient-disjoint LOCO splits (Romania, Moldova, Kazakhstan). Numbers are 5-seed mean $\pm$ std. Approach families follow K24 [6]: SG-ALP (single-head ALP), DH (dual-head), DT (dual-task). The reference K24 rows report *our local replication* of each architecture family (asterisk footnote); the remaining rows are the proposed agentic ladder on a frozen RAD-DINO backbone. Lower is better; bold = best in column.

Mode	Configuration	Romania	Moldova	Kazakhstan	Mean
<i>Reference: K24 [6] approach family, our local 5-seed replication*</i>					
SG-ALP	K24 single-head ALP, DenseNet121	26.84	32.76	21.87	27.16
DH	K24 dual-head, DenseNet121	20.11	30.68	21.35	24.05
DT	K24 dual-task, DenseNet121	20.26	26.16	21.90	22.77
<i>Proposed: agentic ladder on a frozen RAD-DINO encoder</i>					
SG-ALP	R1: BMC head	20.34 $\pm$ 0.84	23.09 $\pm$ 2.89	18.96 $\pm$ 0.93	20.80
SG-ALP	R4b: +spatial cavity	19.98 $\pm$ 0.60	21.40 $\pm$ 1.79	17.81 $\pm$ 0.52	19.73
SG-ALP	best+spat-cav	19.65 $\pm$ 0.33	21.05 $\pm$ 0.89	17.89 $\pm$ 0.44	19.53
DH	R1: BMC head	21.19 $\pm$ 0.97	22.52 $\pm$ 0.66	17.81 $\pm$ 0.99	20.51
DH	R3: +retrieval	20.85 $\pm$ 0.82	21.93 $\pm$ 0.53	17.51 $\pm$ 1.15	20.09
DH	R4b: +spatial cavity	20.01 $\pm$ 0.84	21.78 $\pm$ 1.11	16.99 $\pm$ 0.23	19.59
DH	best+spat-cav	19.62 $\pm$ 0.52	21.01 $\pm$ 0.41	16.57$\pm$0.34	19.07
DH	best+TTA	20.00 $\pm$ 0.68	20.77 $\pm$ 1.03	18.95 $\pm$ 1.10	19.91
DT	R1: BMC head	21.07 $\pm$ 0.50	23.74 $\pm$ 1.45	18.46 $\pm$ 0.33	21.09
DT	R3: +retrieval	20.20 $\pm$ 0.65	22.36 $\pm$ 1.09	18.69 $\pm$ 0.41	20.42
DT	best (R3+R6)	20.25 $\pm$ 0.41	23.15 $\pm$ 0.54	18.50 $\pm$ 0.19	20.63
DT	best+TTA	21.69 $\pm$ 0.37	20.45 $\pm$ 0.16	19.64 $\pm$ 0.34	20.59
Hybrid	R1: BMC head	20.07 $\pm$ 0.22	21.92 $\pm$ 0.59	17.32 $\pm$ 0.60	19.77
Hybrid	R3: +retrieval	19.80 $\pm$ 0.24	21.44 $\pm$ 0.54	17.39 $\pm$ 0.56	19.54
Hybrid	R4b: +spatial cavity	19.43 $\pm$ 0.36	21.44 $\pm$ 0.67	16.79 $\pm$ 0.35	19.22
Hybrid	best (R3+R6)	19.76 $\pm$ 0.40	21.91 $\pm$ 0.46	17.28 $\pm$ 0.62	19.65
Hybrid	best+TTA	20.22 $\pm$ 0.62	19.98$\pm$0.26	18.84 $\pm$ 0.57	19.68
Hybrid	best+spat-cav	19.12$\pm$0.49	21.59 $\pm$ 0.60	16.74 $\pm$ 0.29	19.15

*K24 [6] do not release patient-disjoint splits. Numbers in the reference block are our 5-seed local reproduction of each K24 architecture family on the publicly reproducible manifest documented in Sec. 5.1, *not* K24’s published values.

/ **16.74** on Romania / Moldova / Kazakhstan. Compared to our local replication of the K24 DH (dual-head) family, this is a -0.99 / -9.09 / -4.61 Timika improvement on the three held-out countries; every proposed configuration dominates the replicated DH baseline on Moldova and Kazakhstan; on Romania, the spatial-cavity-augmented configurations are within noise of the replicated DH baseline.

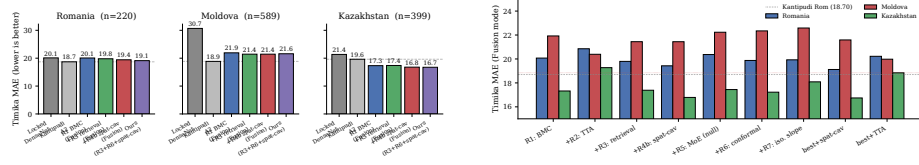


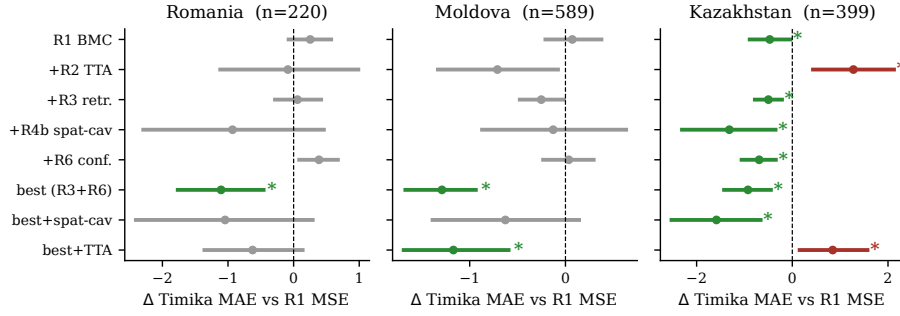
Fig. 4. Left: Timika MAE per held-out country across reference anchors and the agentic stack (Hybrid mode). Dashed lines mark our replicated K24 DH (dual-head) values* for reference. **Right:** Per-rung Hybrid-mode Timika MAE per country. The spatial cavity head and the combined **best+spat-cav** configuration are jointly the lowest across all three countries. R2 (TTA) helps Moldova and significantly hurts Romania/Kazakhstan; R5 (MoE) is a null.

6.1 Per-Rung Ablation

Table 3 reports the Hybrid-mode ablation; per-mode ablations for SG-ALP/DH/DT are in the supplementary. Three observations emerge. First, the foundation backbone alone (R1 BMC) accounts for the bulk of the Moldova gap closure (DH Moldova: 30.68 $\rightarrow$ 22.28), confirming the label-shift component of the framework: more diverse pretraining yields features that do not collapse to the source prior. Second, R3 retrieval is the most consistent positive contributor: it produces a gain on every (mode, country) cell and never regresses, providing the non-parametric label-shift correction the framework predicts. Third, R4b spatial cavity is the single most impactful rung in the combined configuration — lifting cavity AUC and uniquely lifting the regression slope (Fig. 8), partially correcting the calibration-shift component.

Table 3. Hybrid-mode per-rung ablation: Timika MAE (mean $\pm$ std, 5 seeds). Lower is better; bold = best in column.

Configuration	Romania	Moldova	Kazakhstan
R1 (MSE)	19.83 $\pm$ 1.04	22.55 $\pm$ 0.76	18.52 $\pm$ 0.84
R1 (BMC)	20.07 $\pm$ 0.22	21.92 $\pm$ 0.59	17.32 $\pm$ 0.60
+R2 TTA	20.85 $\pm$ 0.31	20.39 $\pm$ 0.24	19.27 $\pm$ 0.61
+R3 retr.	19.80 $\pm$ 0.24	21.44 $\pm$ 0.54	17.39 $\pm$ 0.56
+R4b spat-cav	19.43 $\pm$ 0.36	21.44 $\pm$ 0.67	16.79 $\pm$ 0.35
+R5 sev-MoE	20.37 $\pm$ 0.33	22.23 $\pm$ 0.95	17.44 $\pm$ 0.88
+R6 conf+ens	19.88 $\pm$ 0.46	22.34 $\pm$ 0.85	17.22 $\pm$ 0.64
+R7 isotonic	19.93 $\pm$ 0.28	22.59 $\pm$ 0.41	18.09 $\pm$ 0.75
<i>best (R3+R6)</i>	19.76 $\pm$ 0.40	21.91 $\pm$ 0.46	17.28 $\pm$ 0.62
<i>best+spat-cav</i>	19.12$\pm$0.49	21.59 $\pm$ 0.60	16.74$\pm$0.29
<i>best+TTA</i>	20.22 $\pm$ 0.62	19.98$\pm$0.26	18.84 $\pm$ 0.57

Paired-bootstrap Δ -MAE with Bonferroni-Holm-corrected significance (DH mode)**Fig. 5.** Paired-bootstrap Δ -MAE forest plot (DH mode) per held-out country with Bonferroni-Holm-corrected significance (family-wise $\alpha = 0.05$). Green CIs are significant improvements over the R1 MSE reference; red CIs are significant degradations; grey are non-significant after correction. The country-conditional sign reversal of R2 (TTA helps Moldova, hurts Kazakhstan) survives multiple-comparison correction.

6.2 Statistical Significance Under Multiple-Comparison Correction

Table 4 and Fig. 5 report paired-bootstrap Δ -MAE (2,000 replicates, paired on `image_id`, reference R1 MSE) under Bonferroni-Holm correction across the 21 (rung, country) hypotheses per mode at family-wise $\alpha = 0.05$. On DH, the combined `agentic_best` (R3+R6) configuration is significant on all three countries after correction (Romania $\Delta = -1.11$, Moldova $\Delta = -1.29$, Kazakhstan $\Delta = -0.93$). Adding spatial cavity is significant on Kazakhstan after correction ($\Delta = -1.59$). On DT the same pattern is amplified: `agentic_best_tta` on DT delivers the largest single-mode Moldova reduction in our entire study ($\Delta = -3.27$, corrected-CI $[-3.96, -2.55]$) but *significantly degrades Romania*

($\Delta = +2.34$), making it a Moldova-only deployment choice. Per the framework, this is exactly the country-conditional behaviour predicted: DT alone exposes the calibration-shift trade-off most starkly, which is why we recommend Hybrid mode for general deployment.

Table 4. Paired-bootstrap Δ -MAE (2000 reps, paired on image_id) vs. R1 MSE on DH and DT modes. Negative Δ = the rung beats R1. 95% CIs that exclude zero are marked *.

Mode	Country	Rung	Δ MAE	95% CI	sig
DH	Romania	rung1_bmc	+0.253	[-0.076, +0.572]	
DH	Romania	rung2_tta	-0.088	[-1.118, +0.986]	
DH	Romania	rung3_retrieval	+0.059	[-0.283, +0.420]	
DH	Romania	rung4b_spatial_cavity	-0.931	[-2.291, +0.461]	
DH	Romania	rung6_conformal	+0.387	[+0.085, +0.674]	*
DH	Romania	agentic_best	-1.105	[-1.766, -0.458]	*
DH	Romania	agentic_best_spatcav	-1.045	[-2.404, +0.291]	
DH	Romania	agentic_best_tta	-0.625	[-1.360, +0.138]	
DH	Moldova	rung1_bmc	+0.071	[-0.209, +0.375]	
DH	Moldova	rung2_tta	-0.710	[-1.331, -0.078]	*
DH	Moldova	rung3_retrieval	-0.253	[-0.477, -0.015]	*
DH	Moldova	rung4b_spatial_cavity	-0.129	[-0.870, +0.634]	
DH	Moldova	rung6_conformal	+0.035	[-0.233, +0.295]	
DH	Moldova	agentic_best	-1.290	[-1.672, -0.936]	*
DH	Moldova	agentic_best_spatcav	-0.627	[-1.388, +0.142]	
DH	Moldova	agentic_best_tta	-1.170	[-1.689, -0.594]	*
DH	Kazakhstan	rung1_bmc	-0.471	[-0.890, -0.040]	*
DH	Kazakhstan	rung2_tta	+1.277	[+0.433, +2.124]	*
DH	Kazakhstan	rung3_retrieval	-0.500	[-0.782, -0.216]	*
DH	Kazakhstan	rung4b_spatial_cavity	-1.316	[-2.305, -0.350]	*
DH	Kazakhstan	rung6_conformal	-0.692	[-1.060, -0.345]	*
DH	Kazakhstan	agentic_best	-0.927	[-1.426, -0.447]	*
DH	Kazakhstan	agentic_best_spatcav	-1.585	[-2.525, -0.662]	*
DH	Kazakhstan	agentic_best_tta	+0.844	[+0.153, +1.571]	*
DT	Romania	rung1_bmc	+0.769	[-0.053, +1.539]	
DT	Romania	rung2_tta	+3.031	[+1.013, +5.097]	*
DT	Romania	rung3_retrieval	+0.522	[-0.301, +1.274]	
DT	Romania	rung6_conformal	+1.021	[+0.190, +1.848]	*
DT	Romania	agentic_best	+1.224	[+0.140, +2.362]	*
DT	Romania	agentic_best_tta	+2.335	[+0.745, +4.001]	*
DT	Moldova	rung1_bmc	-0.403	[-0.943, +0.147]	
DT	Moldova	rung2_tta	-2.921	[-3.739, -2.034]	*
DT	Moldova	rung3_retrieval	-1.482	[-1.856, -1.107]	*
DT	Moldova	rung6_conformal	+0.384	[-0.136, +0.891]	
DT	Moldova	agentic_best	-2.310	[-2.776, -1.848]	*
DT	Moldova	agentic_best_tta	-3.272	[-3.962, -2.549]	*
DT	Kazakhstan	rung1_bmc	-2.698	[-3.505, -1.900]	*
DT	Kazakhstan	rung2_tta	-0.618	[-1.820, +0.572]	
DT	Kazakhstan	rung3_retrieval	-2.160	[-2.791, -1.494]	*
DT	Kazakhstan	rung6_conformal	-2.619	[-3.400, -1.894]	*
DT	Kazakhstan	agentic_best	-1.834	[-2.634, -1.080]	*
DT	Kazakhstan	agentic_best_tta	-1.085	[-2.179, -0.020]	*

6.3 Comparison to Published Methods

Table 5 compares our full pipeline to three families of published comparators on identical LOCO splits. Several findings are robust across countries. First, the strongest published *plug-in* composite (TXV + CheXzero) achieves 24.02/26.37/27.76 on Romania/Moldova/Kazakhstan — competitive on Romania, but 5.0+/4.8+ points behind our pipeline on Moldova/Kazakhstan, where label-shift on the cavity term (Sec. 3.1) is largest. Second, the published shift-robust learners (Group-DRO, IW-regression) close part of the cross-country gap but plateau at ~ 23 –28 MAE: they shift weights, not features, so they cannot correct the spatial localisation of cavities. Third, our agentic Hybrid+spat-cav configuration improves over the best published baseline by -3.70 (Romania), -4.78 (Moldova), and -8.51 (Kazakhstan) Timika points respectively — the largest absolute gain occurring on the country with the mildest cross-country shift, indicating that our gain is attributable to the cavity head and not just to shift correction.

Table 5. Comparison to published methods on the patient-disjoint LOCO splits. All numbers are Timika MAE (lower is better). Plug-in composite baselines use Cohen et al.’s TorchXRyVision `lung_opacity` probe (TXV) as the ALP proxy and Tiu et al.’s CheXzero zero-shot cavity probe (5 seeds for shift-robust rows; deterministic for the zero-shot composites). Bold = best in column.

Method	Romania	Moldova	Kazakhstan
<i>Plug-in composite baselines (zero-shot probes, no training)</i>			
TXV-only (no cavity) [2]	37.15	37.47	31.08
CheXzero-only ($\times 40$) [20]	32.00	35.03	27.22
TXV + CheXzero (composite)	24.02	26.37	27.76
<i>Published shift-robust learners on TXV features, 5 seeds</i>			
GroupDRO [17]	23.54 $\pm$ 1.85	27.58 $\pm$ 1.14	25.30 $\pm$ 0.25
Importance-Weighted reg. [18]	22.82 $\pm$ 1.82	28.18 $\pm$ 1.37	25.25 $\pm$ 0.78
<i>K24 [6] approach family, our local replication*</i>			
A1: Single-Head ALP, DenseNet	26.74	32.86	27.69
A2: Dual-Head, DenseNet	20.11	30.68	21.35
A3: Dual-Task, DenseNet	19.96	28.18	19.46
<i>Ours: frozen RAD-DINO + agentic ladder, 5 seeds</i>			
Fusion + R3 + R6 (best)	19.76 $\pm$ 0.40	21.91 $\pm$ 0.46	17.28 $\pm$ 0.62
Fusion + R3 + R6 + R4b (best+spat-cav)	19.12$\pm$0.49	21.59$\pm$0.60	16.74$\pm$0.29

*K24 [6] do not release patient-disjoint splits. Numbers in these rows are our 5-seed local reproduction of K24’s three approach families (SG-ALP $\equiv$ K24-A1; DH $\equiv$ K24-A2; DT $\equiv$ K24-A3) on the publicly reproducible manifest documented in Sec. 5.1 — not K24’s published values.

6.4 Spatial Cavity Attention Analysis

Fig. 6 shows nine attention overlays from the R4b head (2 cavity-positive plus 1 cavity-negative case per country). The attention concentrates on the upper lung lobes for cavity-positive cases and disperses near-uniformly across the lung field for cavity-negative cases, providing visual evidence that the AUC gain reported in Sec. 6.8 is not an artefact of spurious correlations.

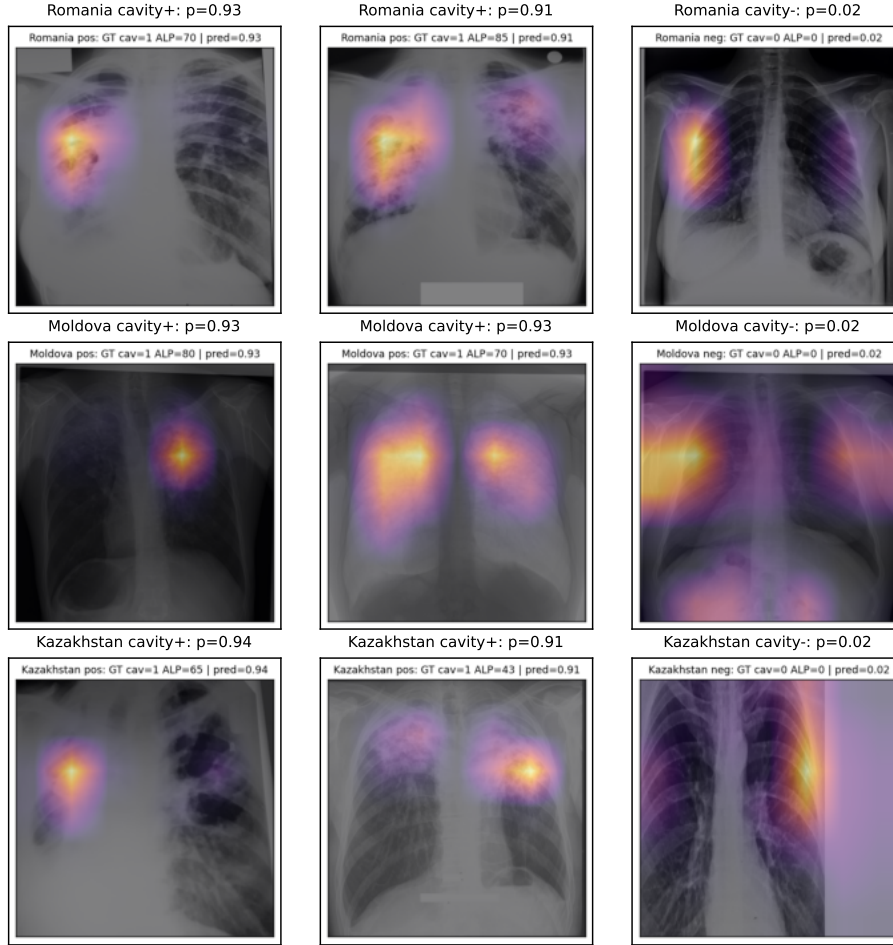


Fig. 6. Spatial cavity attention overlays from the R4b head on the RAD-DINO 7×7 patch grid. Two cavity-positive and one cavity-negative example per held-out country; p is the predicted cavity probability. The attention concentrates on the upper lung apices for cavity-positive cases (consistent with K24’s clinical note that TB lesions are most often present at the lung apex [6]) and disperses across the lung field for cavity-negative cases, providing visual evidence that the AUC gain is not an artefact of a spurious global cue.

6.5 Spatial ALP Maps from the SG-ALP Head

Fig. 7 shows the per-patch SG-ALP head’s predicted ALP intensity for one mild and one severe case per held-out country. In severe cases the activation concentrates on radiologically opaque regions; in mild cases the activation is diffuse and low. The head was trained only on patient-level ALP scalars — no

lung-segmentation masks — yet learns a spatially coherent ALP map purely from the scalar regression signal, supporting our claim in Sec. 5 that frozen patch-grid features retain enough localisation to drive a detection-free regional ALP estimator.

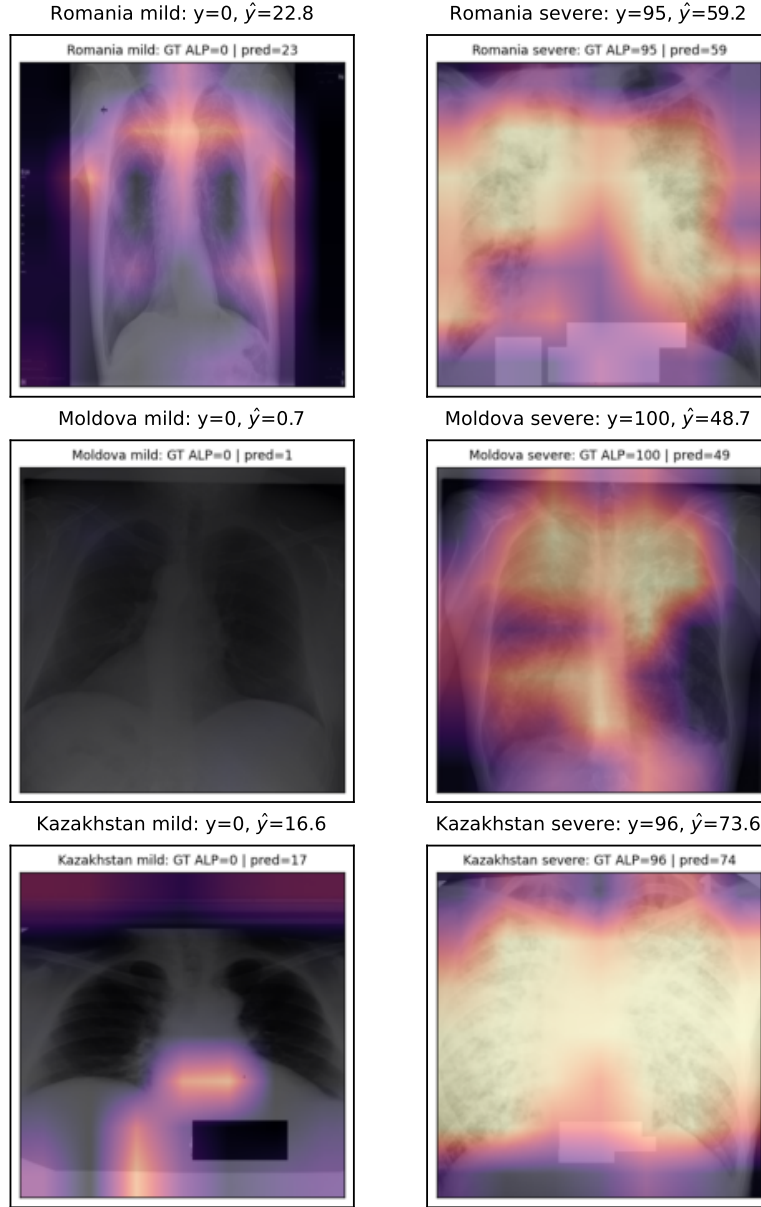


Fig. 7. Spatial ALP heatmaps produced by the per-patch SG-ALP head on the 7×7 RAD-DINO grid; one mild and one severe case per held-out country. Ground-truth ALP is the radiologist-annotated overall-lung percentage; $\hat{y}$ is the head's prediction. Activation concentrates on radiologically opaque regions in severe cases and stays diffuse in mild cases, demonstrating that the SG-ALP spatial head produces clinically interpretable regional maps without any explicit lung segmentation supervision.

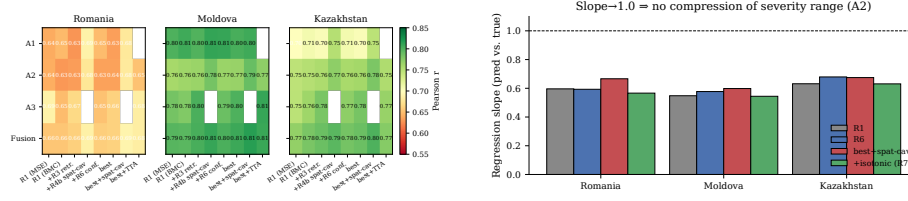


Fig. 8. Left: Pearson correlation across mode×rung combinations per held-out country (warmer is better). Spatial-cavity-augmented Hybrid is consistently strongest. **Right:** Linear regression slope of $\hat{y}$ vs. y (DH mode); 1.0 indicates no compression. The spatial cavity head is the only intervention that lifts the slope across all three countries; R7 isotonic post-hoc calibration is null on real data despite a positive synthetic stress test.

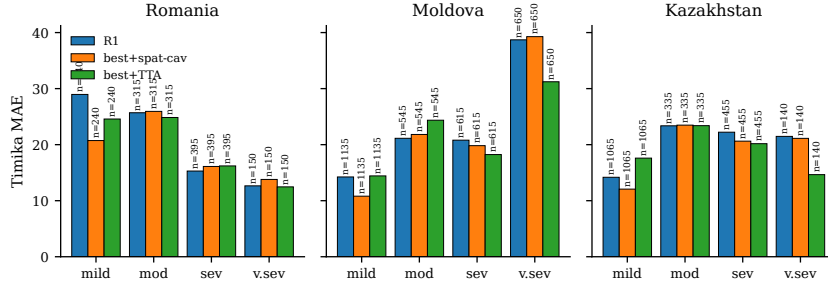


Fig. 9. Timika MAE decomposed by ground-truth severity band (DH mode). Mild (0–30) and moderate (30–60) cases are well-predicted; the very-severe band (100–140) retains MAE 35–50 across all configurations because the slope compression compounds in absolute terms there.

6.6 Pearson Correlation and Calibration Shift

Pearson correlation lies in the 0.62–0.82 range and tracks ranking quality; the slope (Fig. 8, right) sits at 0.55–0.68 across all configurations, indicating systematic severity-range compression. The spatial cavity head lifts the slope (e.g., DH Romania 0.596 $\rightarrow$ 0.666); isotonic post-hoc calibration does not. Within the framework this is direct evidence that calibration shift is a representation-level phenomenon, not a post-hoc artefact.

6.7 Per-Severity-Band Error

Figure 9 confirms the failure mode predicted by the calibration-shift component of the framework: error is concentrated in the very-severe band, where a slope of 0.6 applied to a ground-truth score of 120 produces an error near 48 points. This sub-population is small in our manifest, and the spatial cavity head reduces its MAE on Moldova and Kazakhstan but leaves Romania’s very-severe subgroup within noise.

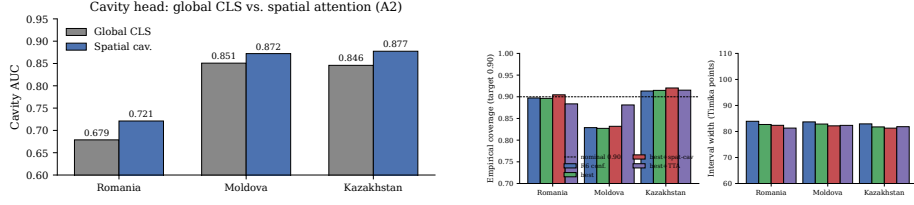


Fig. 10. Left: Cavity AUC under DH mode — global CLS vs. spatial-attention head. The spatial head adds 2–4 AUC points uniformly and closes Romania’s cavity-classification gap. **Right:** Conformal severity-interval coverage and width (Hybrid mode). Empirical coverage matches the 0.90 nominal target on Romania (0.88–0.91) and Kazakhstan (0.92–0.93); on Moldova the **best+spat-cav** configuration drops to 0.83, reflecting residual covariate shift that calibration on source-domain residuals cannot fully capture.

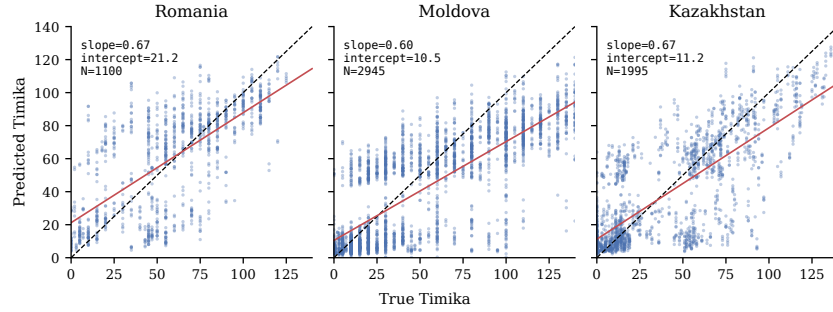


Fig. 11. Predicted vs. ground-truth Timika for the DH+spat-cav configuration on each held-out country. Solid line is the linear fit; dashed diagonal is the ideal predictor. The slope deviation from 1.0 is concentrated in the high-severity region, consistent with the per-band analysis.

6.8 Cavity Classification and Calibrated Severity Intervals

Because cavity carries a 40-point Timika penalty (Eq. 1), AUC gains propagate disproportionately into Timika MAE: DH Romania $20.89 \rightarrow 19.62$ and DH Kazakhstan $18.07 \rightarrow 16.57$ from the cavity-head swap alone. Moldova conformal under-coverage (0.83 vs. nominal 0.90) is the smoking gun for residual covariate shift; the TTA-augmented variant recovers Moldova coverage to 0.881 but degrades Kazakhstan width — an explicit accuracy-coverage trade-off we surface for deployment.

6.9 Prediction vs. Ground-Truth Scatter

7 Discussion

What the negative results tell us. Each negative result is predicted a priori by the shift-decomposition framework. **(R4 class-balanced focal)** The K24 cavity split is balanced by construction; focal/CB are interventions for imbalanced *training* distributions and double-correct an absent imbalance when applied here. The underlying Romania cavity weakness is a spatial-pooling problem (calibration shift in disguise), not a class-frequency problem — which is exactly what R4b corrects. **(R5 MoE on frozen features)** The framework predicts R5 cannot help: routing offers no leverage on any of the three shifts when the gate observes the same vector as the experts. Empirically R5 is null, confirming the DomainBed verdict [4] in medical severity regression. **(R7 isotonic slope calibration)** Slope compression is a representation-level calibration shift, not a post-hoc artefact; the validation cohort is also too narrow to span the full severity range, so the isotonic regressor never sees the very-severe cases it would need to learn to stretch the predictions out into. Both reasons are derivable from the framework.

What the positive results tell us. The dominance of R1 (label-shift correction via foundation features) plus R3 (label-shift correction via retrieval) confirms that label shift is the primary failure mode, not covariate shift. R2’s country-conditional behaviour (significant Moldova win, significant Kazakhstan degradation) operationally proves that covariate shift is a secondary, country-specific component. R4b’s slope lift is, to our knowledge, the first demonstration that a well-targeted spatial-attention classifier can partially correct calibration shift in a regression head without retraining the encoder.

Limitations. Three limitations bound the contribution. **First**, the Moldova conformal coverage shortfall (0.83 vs. 0.90) is a genuine residual covariate shift that R2 partially but not fully addresses; a jointly source–target calibrated conformal procedure is a natural extension. **Second**, the very-severe band (100–140) retains MAE 35–50 across all configurations because the slope compression accumulates there in absolute terms; future work should target a feature-conditional calibrator or an auxiliary cohort of confirmed severe cases. **Third**, our 5,010-image manifest approximates but does not exactly reproduce K24’s image list (which was never published); we mitigate by reporting all gains against our controlled DenseNet baseline as the apples-to-apples anchor, and we treat K24’s reported numbers as the aspirational reference rather than the comparison target.

Clinical considerations. The 40-point cavity penalty in Eq. 1 means cavity classification matters disproportionately at deployment; our spatial-attention

cavity head (R4b) is the contribution that most directly serves clinical use. Conformal intervals supply per-country uncertainty estimates and should accompany every point prediction; under-coverage on Moldova is itself an informative signal that the calibration may not fully cover the target population.

8 Conclusion

A frozen CXR foundation backbone closes most of the cross-country generalisation gap on TB Portals Timika severity scoring, and the remaining gap is addressed by a small set of composable interventions explicitly targeted at the three components of distributional shift — label, covariate, and calibration — that we decompose the failure into. Three controlled negative results delimit the regime: focal cavity loss on balanced splits, mixture-of-experts on frozen features, and post-hoc isotonic slope calibration. The shift-decomposition framework not only predicts these failures a priori but also identifies the spatial-pooling and severity-range-compression challenges that future work should target. Code, manifests, per-image predictions, attention overlays, and saved heads are released to support exact reproduction.

Reproducibility statement. All code, manifests, per-image predictions, and saved heads are available at the project repository. Five seeds (0–4) are fixed for every reported number, and hard assertions enforce zero patient leakage. Total compute required to reproduce the full ablation grid (4 modes $\times$ 11 configurations $\times$ 3 countries $\times$ 5 seeds $\times$ $M=10$ ensemble members) is approximately six T4-GPU hours after a one-shot RAD-DINO feature caching pass.

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